

### DETERMINATION OF STRESSES AND STRAINS FOR UNIFORM BAR

**Aim:** To write and execute a Python program for calculating stress, longitudinal strain and lateral strain for circular and rectangular cross sections using user-defined functions .

**Interactive Development Environment (IDE) Used :** Jupyter Notebook

**Programming Language Used :** Python

**Basic theory .:**

Stress is the internal resisting force per unit area. Longitudinal strain is the ratio of change in length to original length. Lateral strain is the ratio of change in lateral dimension to original dimension. This experiment uses modular programming with user-defined functions and a while loop.

**Equations:**

**Circular Area:**  $A = \pi d^2/4$

Rectangular Area:  $A = b \times t$

Stress:  $\sigma = P/A$

Change in Length ,  $\Delta l = PL/(AE)$

Longitudinal Strain:  $\epsilon = \Delta l / l$

Lateral Strain:  $\epsilon = \Delta d/d$  or  $\Delta b/b$

**Python Concepts used:**

A function is a named block of code that performs a specific task. Functions improve the readability, modularity, and reusability of a program by dividing a large problem into smaller manageable parts. Instead of writing the same code repeatedly, a function can be written once and called whenever required.

A function may:

- accept arguments (parameters) from the calling program,
- perform the required computations,
- optionally return one or more values, or simply perform an action without returning any value.

| Function Type                                                | Function Definition (Template)                                                                                                                                                                    | Function Call                                                            |
|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| 1. Function without Return Value                             | def display():<br>print("Hello Students")                                                                                                                                                         | display()                                                                |
| 2. Function with One Return Value                            | def square(num):<br>result = num * num<br>return result                                                                                                                                           | value = square(5)<br>print(value)                                        |
| 3. Function with Two Return Values                           | def rectangle(length, breadth):<br>area = length * breadth<br>perimeter = 2 * (length + breadth)<br>return area, perimeter                                                                        | a, p = rectangle(10, 5)<br>print("Area =", a)<br>print("Perimeter =", p) |
| 4. Function Used for Displaying Results (Current Experiment) | def display(area, stress, long_strain, lat_strain):<br>print("Area =", area)<br>print("Stress =", stress)<br>print("Longitudinal Strain =", long_strain)<br>print("Lateral Strain =", lat_strain) | display(area, s, long_strain, lat_strain)                                |
| 5. Function                                                  | def stress(load, area):                                                                                                                                                                           | s = stress(P, area)                                                      |

|                                                      |                                                                                                                     |                                                  |
|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| Returning One Value (Current Experiment)             | $\sigma = \text{load} / \text{area}$<br>return $\sigma$                                                             |                                                  |
| 6. Function Returning One Value (Current Experiment) | def longitudinal_strain(load, length, area, E):<br>long_strain = (load * length) / (area * E)<br>return long_strain | long_strain = longitudinal_strain(P, L, area, E) |
| 7. Function Returning One Value (Current Experiment) | def lateral_strain(change, original):<br>lat_strain = change / original<br>return lat_strain                        | lat_strain = lateral_strain(delta_d, d)          |

### Python Program:

```
# Program: Stress and Strain Calculator Using Functions and While Loop
import math

# Function to Display Results
def display(area, stress_value, long_strain, lat_strain):
    print("\n----- RESULTS -----")
    print("Area                =", round(area, 2), "mm2")
    print("Stress                 =", round(stress_value, 2), "N/mm2")
    print("Longitudinal Strain    =", round(long_strain, 6))
    print("Lateral Strain         =", round(lat_strain, 6))

# Function to Calculate Stress
def stress(load, area):
    sigma = load / area
    return sigma

# Function to Calculate Longitudinal Strain
def longitudinal_strain(load, length, area, young_modulus):
    long_strain = (load * length) / (area * young_modulus)
    return long_strain

# Function to Calculate Lateral Strain
def lateral_strain(change, original):
    lat_strain = change / original
    return lat_strain

# Function for Circular Cross Section
def circular_section():
    print("\nCIRCULAR CROSS SECTION")
    d = float(input("Enter Diameter (mm): "))
    L = float(input("Enter Length (mm): "))
    P = float(input("Enter Tensile Load (N): "))
    E = float(input("Enter Young's Modulus (N/mm2): "))
    delta_d = float(input("Enter Change in Diameter (mm): "))
    area = math.pi * d * d / 4
    s = stress(P, area)
    long_strain = longitudinal_strain(P, L, area, E)
    lat_strain = lateral_strain(delta_d, d)
    display(area, s, long_strain, lat_strain)

# Function for Rectangular Cross Section
def rectangular_section():
    print("\nRECTANGULAR CROSS SECTION")
    b = float(input("Enter Width (mm): "))
    t = float(input("Enter Thickness (mm): "))
    L = float(input("Enter Length (mm): "))
    P = float(input("Enter Tensile Load (N): "))
    E = float(input("Enter Young's Modulus (N/mm2): "))
    delta_b = float(input("Enter Change in Width (mm): "))
    area = b * t
    s = stress(P, area)
    long_strain = longitudinal_strain(P, L, area, E)
    lat_strain = lateral_strain(delta_b, b)
    display(area, s, long_strain, lat_strain)
```

```

# Main Program
while True:
    print("\nSTRESS AND STRAIN CALCULATOR")
    print("1. Circular Cross Section")
    print("2. Rectangular Cross Section")
    print("3. Exit")
    choice = int(input("Enter your choice (1, 2 or 3): "))
    if choice == 1:
        circular_section()
    elif choice == 2:
        rectangular_section()
    elif choice == 3:
        print("\nProgram Terminated.")
        break
    else:
        print("\nInvalid Choice.")
        again = input("\nDo you want to calculate another structure? (Y/N): ")
        if again.upper() != "Y":
            print("\nThank You.")
            break

```

### **Sample Input and Output :**

### **Result :**

The Python program was written and executed successfully to calculate stress and strain values.